

9. Bias-Variance Tradeoff

Friday, January 30, 2026 3:04 PM
1/23/2022, updated 2026

Error Decomposition aka Bias-Variance Tradeoff § 5.2 [SS]

Let $h_S \in \text{ERM}_{\mathcal{H}}(S)$

Total Error (true risk) is

$$L_D(h_S) = \underbrace{L_D(h_S) - \min_{h \in \mathcal{H}} L_D(h)}_{\text{estimation error}} + \underbrace{\min_{h \in \mathcal{H}} L_D(h)}_{\text{approximation error}}$$

$$\text{or } L_D(h_S) - \underbrace{\min_{h \in \mathcal{H}} L_D(h)}_{\text{Bayes risk, } \epsilon_{\text{Bayes}}} = \epsilon_{\text{est}} + \tilde{\epsilon}_{\text{approx}}$$

$\tilde{\epsilon}_{\text{approx}} = \epsilon_{\text{approx}} - \epsilon_{\text{Bayes}}$

"excess risk"

ϵ_{est} : error due to using $\hat{L}_S$ instead of L_D . This is what our agnostic PAC bounds cover.

aka generalization. Results like $m = O\left(\frac{\log(|\mathcal{H}|/\delta)}{\epsilon^2}\right)$...

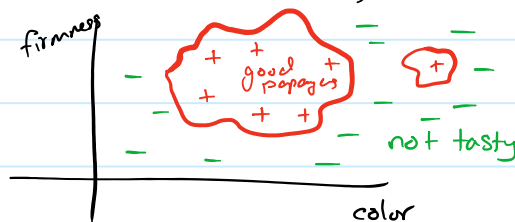
Smaller $|\mathcal{H}|$ is "good": better sample complexity, generalization, lower variance

The intuition is low complexity $\mathcal{H}$ is good

ex: $\mathcal{H} = \{h\}$, $\epsilon_{\text{est}} = 0$
so ERM data independent
 h isn't a r.v.
and we have concentration.

ϵ_{approx} : we've ignored so far. This is more classical, or unknowable a priori.

Ex papayas



hard to analyze
inductive bias/
exper

We're looking to approximate h_{true} by $\underline{h \in \mathcal{H}}$

Ex $\mathcal{H} = \{\text{all polynomials of degree } \leq 100\}$

h_{true} is an arbitrary function (maybe even continuous)

ϵ_{approx} unlikely 0

(p. 2: bias-variance tradeoff)

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$\mathcal{E}_{\text{Bayes}}$

accounts for inherent noise in labels

eg, (pure) PAC: $y = f(x)$ so $\min_h L_D(h) = L_D(f) = 0$

eg, agnostic PAC: $y \sim D(y|x)$

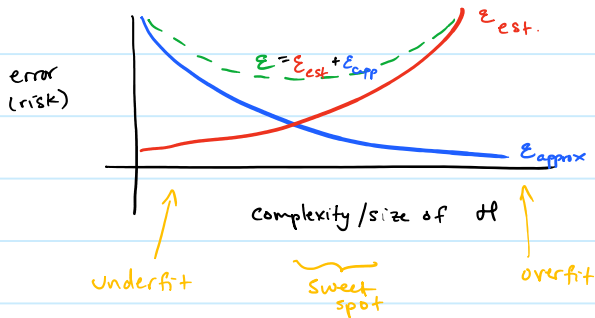
eg $y = f(x) + z$, $z \sim N(0, \sigma^2)$

how close to deterministic
is our model

Inductive bias of X plays
a role

not our focus

(textbook) bias-variance tradeoff

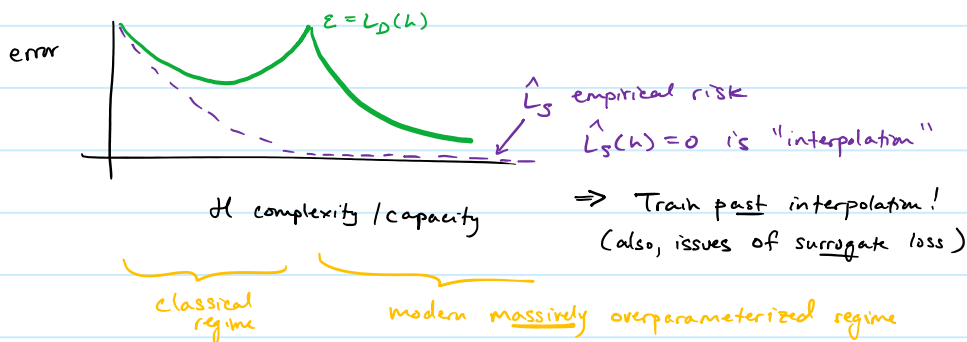


} often compute experimentally

Point: both terms matter, so $|H|$ small not always good
 $\mathcal{E}_{\text{approx}}$ hard to predict a priori

Double-descent

Belkin, Hsu, Ma, Mandal PNAS '19



Classical: "A model with 0 training error is overfit to the training data
and will typically generalize poorly"

(Paz, Hastie, Tibshirani, Friedman

"The Elements of Statistical Learning" 2001)

That can be true, but need not be

- Deep Learning (2013+) empirical results strongly show
best results obtained on massively overparameterized ($\hat{L}_S(h) = 0$)

neural networks

(caveat: how you train matters, eg., $\arg\min_h \hat{L}_S(h)$ is
a big set, and not all ERM sol'n generalize...
but the ones we find via SGD work well)

(p. 3: bias-variance tradeoff)

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- Double descent can show up
Though I'd disagree that it always does

We don't fully understand deep learning yet
many ideas in lit.

That paper suggests

- our notions of complexity not a good measure
- smoothness of h might be better
(already sometimes exploited)
- d larger might allow for smoother functions
(exactly at interpolation threshold is likely not smooth)
- or, w/ many sol'n to choose from, a least-squares
(or other crit.) works well

2026 addendum

- see extra notes on James-Steen estimator
- classical **bias-variance tradeoff** is: Say, mean $\mu := \mathbb{E} \hat{\theta}$

$$\begin{aligned} \text{MSE}(\hat{\theta}) &:= \mathbb{E}(\hat{\theta} - \theta)^2 = \mathbb{E}(\hat{\theta}^2 - 2\theta\hat{\theta} + \theta^2) \\ &= \mathbb{E}(\hat{\theta}^2) - 2\mu\theta + \theta^2 \end{aligned}$$

θ isn't random

$$\text{Claim: } \text{MSE}(\hat{\theta}) = \underbrace{(\mu - \theta)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}(\hat{\theta} - \mu)^2}_{\text{variance}}$$

$$\begin{aligned} &= \underbrace{\mu^2 - 2\mu\theta + \theta^2}_{\text{bias}^2} + \underbrace{\mathbb{E}\hat{\theta}^2 - 2(\mathbb{E}\hat{\theta})\mu + \mu^2}_{\text{variance}} \\ &= \cancel{\mu^2} - 2\mu\theta + \theta^2 + \mathbb{E}\hat{\theta}^2 - \cancel{\mu^2} + \mu^2 \end{aligned}$$

match ✓